

Hands-On: Rendering Lists of Data with Keys in React

In this lab, you will create a simple React component that uses an array of data to render a list of components with unique keys.

Step 1: Create a new React component

Create a new file called ``UserList.js`` and add the following code:

```
import React from 'react';

const users = [
  { id: 1, name: 'Alice' },
  { id: 2, name: 'Bob' },
  { id: 3, name: 'Charlie' },
];

function UserList() {
  return (
    <div>
      <h1>User List</h1>
      {users.map(user => (
        <User key={user.id} user={user} />
      ))}
    </div>
  );
}

function User({ user }) {
  return <div>{user.name}</div>;
}

export default UserList;
```

This component has an array of user data and a ``User`` component that renders the user's name. The ``UserList`` component maps over the array and renders a ``User`` component for each user item with a unique key based on the ``id`` property.

Step 2: Render the component

Create a new file called ``App.js`` and add the following code:

```
import React from 'react';
import UserList from './UserList';

function App() {
  return (
    <div className="App">
      <h1>Rendering Lists of Data with Keys in React</h1>
      <UserList />
    </div>
  );
}
```

```
export default App;
```

This component simply renders the `UserList` component.

Step 3: Run the app

Run the app using `npm start` and test the `UserList` component by verifying that each user item has a unique key and that the user's name is displayed.

Congratulations! You have successfully rendered lists of data with keys in a React component.